ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4 (Q1 struck out – not to be considered)

Question 1 [CROSSED OUT BY STUDENT]

~~*Given: sensor 1280 px wide, pixel pitch $p = 5\ \mu m$ (square), lens $f = 16$ mm, standoff $Z = 400$ mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, $T = 0.1$ s.*~~

~~**(a)** Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.~~

~~(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\ \text{m})/(5\times10^{-6}\ \text{m}) = 3200$. The metres cancel so the anwser is in pixels. **$f_x = 3200$ px**.~~

~~(ii) Width of the sensor $= N\cdot p = 1280 \times 5\ \mu m = 6400\ \mu m = 6.4$ mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s\cdot Z/f = (6.4\ \text{mm})(400\ \text{mm})/(16\ \text{mm}) = 160$ mm. **FOV = 160 mm**.~~

~~Ground sampling distance $= FOV/N = 160\ \text{mm} / 1280\ \text{px} = 0.125$ mm/px. Cross-check with the pixel based formula $Z/f_x = 400\ \text{mm} / 3200\ \text{px} = 0.125$ mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.~~

~~**(b)** Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisaton sets the limit.~~

~~Voltage step: $1\ \text{LSB} = V_{FS}/2^N = 5\ \text{V}/4096 = 1.2207\times10^{-3}\ \text{V} = 1.2207$ mV.~~

~~The sensor maps force to voltage by $V = S\cdot F$ with $S = 10$ mV/N, so $\Delta F_{min} = \Delta V/S = 1.2207\ \text{mV} / (10\ \text{mV/N}) = 0.12207$ N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)~~

~~**(c)** With u held constant over $[kT, (k+1)T]$ (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since $u(kT)$ is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.~~

~~Substituting $A = -2$, $B = 1$, $T = 0.1$ s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.~~

~~Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.~~

~~**$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$**~~

~~*(Cancelled - please do not evaluate this question.)*~~

Question 2

Given: $J = 0.02$ kg·m², $b = 0.05$ N·m·s/rad, $m = 1$ kg, $\ell = 0.25$ m, $g = 9.81$ m/s². So $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so **$|\tau_0| \le mg\ell = 2.4525$ N·m**. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) The equilibrum at $\theta_e = 60°$ needs the torqe $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

$A = [0, 1; -61.31, -2.5]$, and since $\partial f_2/\partial\tau = 1/J = 50$, **$B = [0; 50]$**.

Check: the characterstic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) At steady state $\dot{\theta} = 0$ and $\ddot{\theta} = 0$.

A PD controler is stabel and its job is to drive $(\theta_d - \theta)$ to zero, so $\theta_{ss} = \theta_d = 90°$. The gravity term only affects the transient. Hence the steady-state error is **0°** and the joint does come to rest at θ_d.

(d) Steady state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: stedy-state error is zero**.

Pols: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell\cos\theta_d)\,\delta\theta = 0$, characteristic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell\cos\theta_d$. After compensation the added term $mg\ell\sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell\cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because $\cos 90° = 0$; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3×0.02... i.e. it would move the poles.

Question 3

Given: $A = [0, 1; 0, -2]$, $B = [0; 1]$, $C = [1, 0]$, $f_s = 1$ kHz so the sample time is $T = 1/f_s = 1$ ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency ($\approx \omega$ for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) $\omega_c = 20$ rad/s. Tustin with T = 0.001: $a = \omega_c T/2 = 0.01$.

$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 - a)/(1 + a)]\,y_{k-1}$ = **$0.00990\,(u_k + u_{k-1}) + 0.98020\,y_{k-1}$**.
DC gain: $2(0.0099)/(1 - 0.9802) = 0.0198/0.0198 = 1$ ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\,s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desired polynomial is $(s + 10)^2 = s^2 + 20\,s + 100$.
Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

L = [18, 64]^T. Verification: A − LC = [−18, 1; −64, −2] has trace −20 and determinant (−18)(−2) − (1)(−64) = 36 + 64 = 100, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Advantage of the obsever that no fliter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimat therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the fliter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or acuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

Sampling at 1 kHz is fast compared with the 10 rad/s observer bandwidth, so discretisaton effects are small.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\,r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain $= bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) With the actual b = 0.75 and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 =$ **0.8**.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 =$ **20%**. A common mistake would be to say the output falls by the same 25% as the acuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Substitutting u and differentiating: the integral term enters with a minus sign, so the characterstic polynomial is $s^3 + (a + bk_2)s^2 + (c + bk_1)s - bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

So $2 + k_2 = 8 \Rightarrow k_2 = 6$; $6 + k_1 = 24 \Rightarrow k_1 = 18$; $-K_I = 32 \Rightarrow$ **$K_I = -32$**.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. **$\theta_{ss} = r$**.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the acuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrator.

Remark: with integral action the exact value of N_{ff} matters less because the integrater enforces tracking.